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RESEARCH ARTICLE

Excitation-Contraction Coupling, Electrophysiology, and Arrhythmias

Interpretation of field and LEAP potentials recorded from cardiomyocyte monolayers

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Abstract

Multielectrode arrays (MEAs) are the method of choice for electrophysiological characterization of cardiomyocyte monolayers. The field potentials recorded using an MEA are like extracellular electrograms recorded from the myocardium using conventional electrodes. Nevertheless, different criteria are used to interpret field potentials and extracellular electrograms, which hamper correct interpretation and translation to the patient. To validate the criteria for interpretation of field potentials, we used neonatal rat cardiomyocytes to generate monolayers. We recorded field potentials using an MEA and simultaneously recorded action potentials using sharp microelectrodes. In parallel, we recreated our experimental setting in silico and performed simulations. We show that the amplitude of the local RS complex of a field potential correlated with conduction velocity in silico but not in vitro. The peak time of the T wave in field potentials exhibited a strong correlation with APD_{90} while the steepest upslope correlated well with APD_{50}. However, this relationship only holds when the T wave displayed a biphasic pattern. Next, we simulated local extracellular action potentials (LEAPs). The shape of the LEAP differed markedly from the shape of the local action potential, but the final duration of the LEAP coincided with APD_{90}. Criteria for interpretation of extracellular electrograms should be applied to field potentials. This will provide a strong basis for the analysis of heterogeneity in conduction velocity and repolarization in cultured monolayers of cardiomyocytes. Finally, a LEAP is not a recording of the local action potential but is generated by intracellular current provided by neighboring cardiomyocytes and is superior to field potential duration in estimating APD_{90}.

NEW & NOTEWORTHY We present a physiological basis for the interpretation of multielectrode array-derived, extracellular, electrical signals.

field potential; IPS; LEAP; monolayer

INTRODUCTION

Cultured monolayers are a widely used model to study electrical heart disease in vitro (1, 2). They can be engineered by growing ventricular cardiomyocytes from neonatal rats (NRVMs) or inducible pluripotent stem cells (IPS) (3). The preparation is suitable for high-throughput studies and has been applied for not only drug screening or genetic interventions (4) but also mechanistic studies. Electrophysiological characterization of these monolayers is based on noninvasive recordings of extracellular potentials and is often performed in a semiautomated manner using micro- or multielectrode arrays (MEAs) (5, 6). In this setting, the extracellular potentials recorded using an MEA are referred to as field potentials. Although recording of extracellular potentials is widely used in clinical cardiology and interpretation of these local electrograms has been the subject of research since the 1970s (7–13, 37), this body of evidence is rarely used when interpreting field potentials recorded from monolayers (14, 15). Applying established knowledge will improve signal interpretation of extracellular potentials derived from monolayers, promote translation of results to the intact heart, and ultimately to cardiac pathology and patients with arrhythmias.

Here, we link the morphology of the extracellular potential derived from a monolayer to the underlying local action potential. We measured field potentials and recorded trans-

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membrane potentials from a monolayer of cultured NRVMs. In parallel, we performed computer simulations to replicate the recorded signals and systematically and quantitatively compared the extracellular signal morphology with the directly underlying action potential shape. In particular, we correlated the commonly used surrogate measures of action potential duration, derived from the local T wave of a field potential, with the actual duration of the local action potential. We also addressed the mechanism underlying the generation of a local extracellular action potential (LEAP) and correlated it with local true action potentials.

METHODS

Animal Ethics

Our methods adhere to the Animal Research: Reporting of In Vivo Experiments (ARRIVE) Guidelines. All animal experiments were approved by the local Animal Experiments Committee (Academic Medical Center, University of Amsterdam) and carried out in compliance with the *Guide for the Care and Use of Laboratory Animals* and in accordance with national and institutional guidelines.

Generation of Monolayers

The isolation of NRVMs was performed as previously described (2). Briefly, hearts were rapidly excised from 0- to 2-day-old male and female Wistar rats. The ventricles were cut into pieces and dissociated with trypsin (1 mg/mL; Sigma) and collagenase type 2 (1 mg/mL, Worthington Vollenhove, The Netherlands, 230 U/mg). Cells were collected and resuspended in culture medium (M199 medium, Gibco) supplemented with 10% heat-inactivated fetal bovine serum (FBS, Gibco), 1% HEPES (Gibco), 5,000 U/L penicillin-G (Sigma), 2 mg/L vitamin B12 (Sigma-Aldrich), 3.5 g/L glucose, 1% nonessential amino acids (Gibco), and 1% L-glutamine (Gibco). The cell suspension was preplated to minimize fibroblast contamination. The remaining myocardial cells were plated on microelectrode arrays (MEAs, Multi-Channel Systems, Reutlingen, Germany) coated with fibronectin (125 μg/mL BD Biosciences, Breda, The Netherlands) at a density of 1.4×10^5 cells/cm^2. NRVMs were cultured at 37°C and 5% CO_2. After 24 h of culture in medium supplemented with 10% FBS, cells were cultured in DMEM (Gibco) supplemented with 10% FBS (Gibco), 1% HEPES (Gibco), 5,000 U/L penicillin-G (Sigma), 2 mg/L vitamin B12 (Sigma-Aldrich), 4.5 g/L glucose, 1% nonessential amino acids (Gibco), and 1% L-glutamine (Gibco).

Electrical Mapping and Microelectrode Measurements

Electrical mapping was performed on 23 NRVM monolayers (8 to address conduction velocity and 15 to address repolarization) on *day 4* after isolation as previously described (2). Briefly, 10 min before measurements, NRVM culture media was replaced by a modified Tyrode's solution (36.5°C) containing (in mM) 140 NaCl, 5.4 KCl, 1.8 $CaCl_2$, 1.0 $MgCl_2$, 5.5 glucose, and 5.0 HEPES (pH 7.4, adjusted with NaOH). NRVMs cultured on MEAs (a matrix of 60 electrodes, electrode diameter of 100 μm with an interelectrode distance of 700 μm, in the center of a tissue bath of 15 mm diameter) were stimulated using a bipolar extracellular stimulus electrode (twice diastolic stimulation threshold, 1–2-ms pulse). The reference electrode was placed in the bath without touching the cardiomyocytes. From field potentials, activation time (AT) was determined as the interval from the stimulation artifact to the time of the minimum derivative of the local RS and used to construct activation maps. Conduction velocity was determined along lines perpendicular to isochronal lines by dividing the distance by the difference in local AT, e.g., the vector method (16). Local conduction velocity was determined by dividing the average absolute time difference between an electrode and the surrounding electrodes by the interelectrode distance. Following electrical mapping, spontaneous action potentials were measured in NRVM monolayers using glass pipette microelectrodes (Harvard apparatus GC100F-10). The transmembrane potentials were recorded within 1 mm from an extracellular electrode. The transmembrane potentials were preamplified and recorded simultaneously with the field potentials using the same amplifier. These microelectrodes were filled with 3 M KCl, typical tip resistance was 15–25 MΩ. An AgCl-coated silver wire was used as a reference electrode, placed in the bath and without contact with the monolayer. Maximum diastolic potential and maximum action potential (AP) upstroke velocity (V_{max}) were calculated. AP duration was determined at 20, 50, and 80% (APD_{20}, APD_{50}, APD_{80}) of the AP repolarization amplitude starting from the peak of the action potential. All signal analyses were performed using a custom-made data analysis program written in Matlab 2006 b (MathWorks, Natick, MA).

LEAP Recordings from Human Induced Pluripotent Stem Cell-Derived Cardiomyocytes

Human induced pluripotent stem cell-derived cardiomyocytes (iPSCs) were differentiated toward ventricular and atrial cardiomyocytes as described previously (17). At *day 16* cardiomyocytes were dissociated using 1× TryPLE Select and plated on Matrigel-coated MEA Plates (24-well Lumos Plate) at a density of 4.0×10^3 per coverslip. Cells were measured 9 days postseeding. Plates were allowed to equilibrate in the MEA for 30 min before measurements. We recorded potentials from 12 monolayers for 2 min for each plate using AxIS Navigator 2.0.4 (Axion BioSystems) and analyzed with the Cardiac Analysis Tool 2.2.7 (Axion BioSystems).

Computer Simulations

All simulations were performed using the openCARP cardiac electrophysiology simulator (18, 19). The simulated tissue was a round two-dimensional (2-D) sheet with a diameter of 24 mm and an MEA placed at its center. The elements under the MEA had edge lengths of 100 μm, whereas elements outside this region had edge lengths of 300 μm. Membrane kinetics were represented by the NRVM model of Wang and Sobie (20), the O'Hara model for the human (21) ventricular action potential, and the Courtemanche model for the human atrial action potential (22). To simulate regions under the LEAP electrode, a membrane model with a fixed transmembrane voltage of −50 mV and fixed membrane resistance of 1 kΩ/cm^2 was implemented over a 100 × 100 μm^2 area (23). A monodomain formulation was used with a time step of 25 μs, and results were output every 100 μs for analysis. The space constant was 0.865 mm. As a monodomain formulation was

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used, a single tissue conductivity value characterized intercellular coupling (24). Extracellular potentials were computed by a monopole approximation calculating the current leaving the cells, and assuming electrodes were 100 μm above the surface of the tissue (24).

RESULTS

Fitting the In Silico Model to Experimental Values

The experimental procedure is schematically depicted in Fig. 1*A*. To define the physiological range of conduction velocities and action potential durations in the in silico model, we first recorded field potentials from cultured NRVMs from which we derived activation maps (Fig. 1, *B* and *C*). A single transmembrane action potential was recorded within the field of 60 extracellular recording electrodes (Fig. 1*C*).

We then used an in silico model based on the mathematical model of neonatal rat ventricular action potential. Simulated propagation and repolarization are seen in Fig. 1, *D* and *E*. Without adjustments, the conduction velocity was much faster than observed during experimental measurements (±1.4 vs. 0.2 m/s). The AP duration was much shorter in the simulated than measured action potentials (77 vs. 160 ± 9 ms). Therefore, we adjusted different conductivities in the in silico model (Table 1) to obtain action potential durations that were similar to the experimentally measured durations. Moreover, the intercellular coupling was reduced by 75% to obtain conduction velocities comparable to those in the experimental setting.

Influence of Conduction Velocity on the Shape of Field Potential

To determine the influence of conduction velocity on the shape of the field potential in different physiological and pathological conditions, we adjusted the conductance of the fast sodium current (g_{Na}) to generate a wide range of conduction velocities below and above baseline conditions.

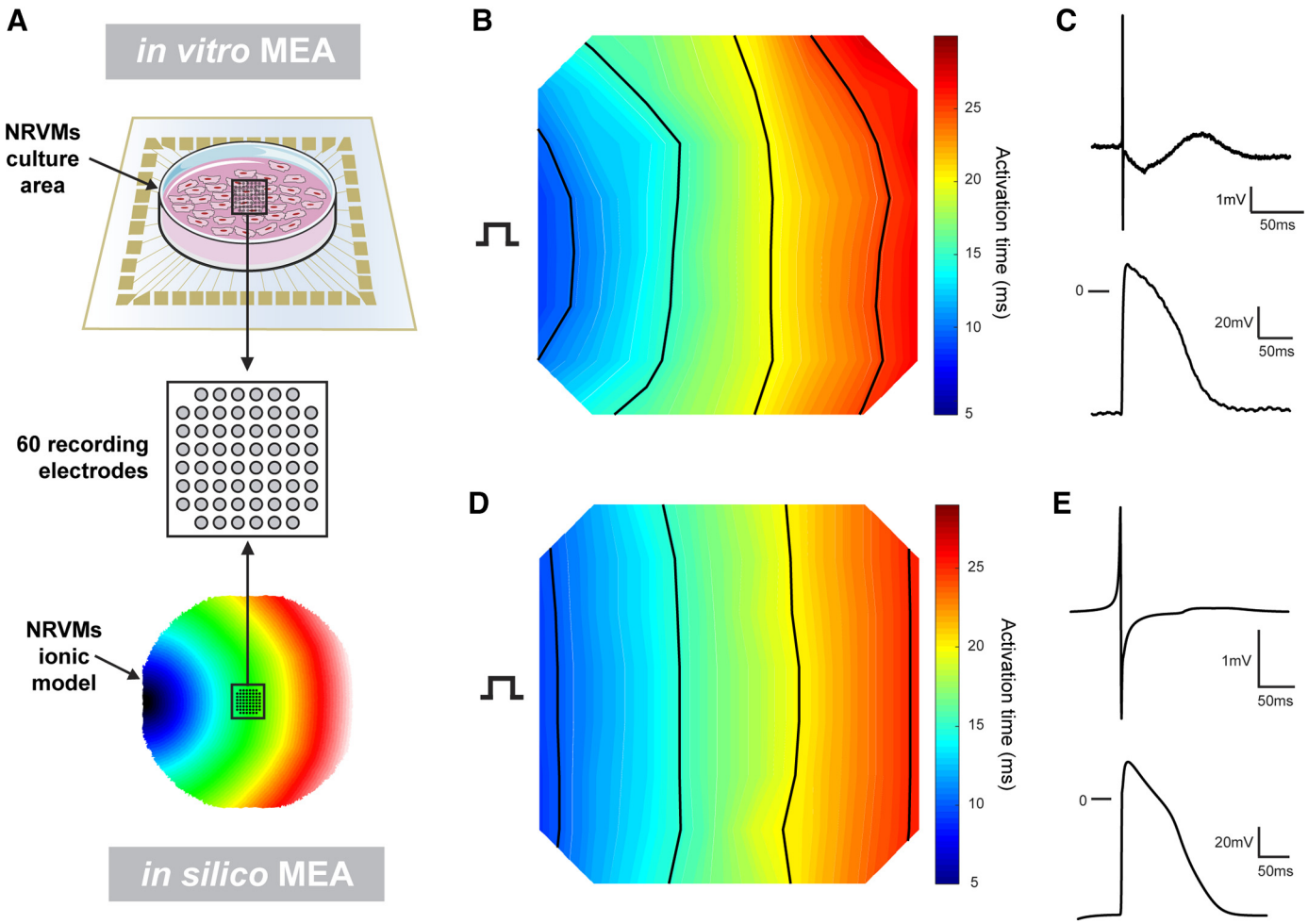


Figure 1. In silico multielectrode array (MEA) model of neonatal rat ventricular myocytes (NRVMs). *A*: study design: neonatal rat ventricular myocytes (NRVMs) were cultured on a multielectrode array (MEA, 60 recording electrodes) and electrical activity was measured on *day 4*. Simulations were performed using an NRVM ionic model with similar cell density and electrode grid characteristics. *B*: activation map from the 60 electrodes grid on which neonatal rat ventricular myocytes were cultured. Local activation time was defined as the time of the maximum downslope of the field potential. Activation times according to the scale on the *right*. *C*: field potential was recorded from one of the electrodes of the MEA (*C*, *top*) and action potential (*C*, *bottom*) was obtained by microelectrode recording of cultured NRVMs in the same MEA. *D*: activation map from the electrode grid area of the in silico model of NRVMs, field potential from one electrode (*E*, *top*) and transmembrane recording (*E*, *bottom*) from the electrode location. Isochrones at 5 ms. Activation times according to the scale on the *right*.

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Table 1. *Summary of the changes applied to the Wang ionic model*

Ionic Current	Modifier, %
g_{K1}	−50
g_{Kr}	−75
g_{Ktof}	−50
$I_{Ca,L_slowdown}$	+10
g_{Kss}	−90
g_{Ks}	−90
g_{Na}	+15

g_{K1}, inward rectifier potassium current conductance; g_{Kr}, rapid-delayed rectifier potassium current conductance; g_{Ktof}, fast-inactivating transient outward potassium current conductance, $I_{ca,L_slowdown}$, L-type calcium current; g_{Kss}, steady-state potassium current conductance; g_{Ks}, slow-delayed rectifier potassium current conductance; g_{Na}, fast sodium current conductance.

We performed 49 simulations in which we modified g_{Na} in each simulation by steps of 0.1-fold, ranging from 0.35 to 5.15 of the baseline g_{Na} value. Figure 2*A* shows field potentials at normal g_{Na} (*1.05) in black and at high g_{Na} (*5.15) in blue. The amplitude of the field potential (RS complex) was much larger at high g_{Na} than at normal g_{Na} (5.7 vs. 3.7 mV, respectively). Figure 2*B* shows two examples of activation maps obtained in simulations where g_{Na} was normal (left) and high g_{Na} (right), leading to conduction velocities of 30 and 47 cm/s, respectively. Both the amplitude of the field potential and the dV/dt_{min} of the downward slope of the R wave correlated positively with the conduction velocity ($r = 0.96$ and $r = 0.90$, respectively, $P < 0.001$) (Fig. 2*C*). These results indicate that both the amplitude of the field potential and the dV/dt_{min} of the downslope of the RS complex correctly estimate changes in conduction velocity in the lower range of conduction velocities, in silico. For conduction velocities in the normal range (above 30 cm/s), these correlations are lost.

To compare in silico results with experimental recordings, we measured conduction velocity at a basal cycle length (S1) of 600 ms and after premature stimulation 10 ms above the refractory period (S2). We calculated the local conduction velocity at two locations marked A and B in Fig. 2, *D* and *E*. The conduction velocity varied between 6 and 26 cm/s across the monolayers. Both the amplitude of the field potential and the dV/dt_{min} of the downward slope of the R wave correlated positively with conduction velocity. However, the Pearson coefficient was lower than in the computer simulations ($r = 0.96$ vs. 0.22 and $r = 0.9$ vs. 0.29, respectively), and, importantly, these correlations did not reach statistical significance. Local conduction velocity did not change after premature stimulation (Fig. 2*G*), whereas field potential amplitude and dV/dt_{min} became significantly smaller (Fig. 2, *H* and *J*). The latter suggests that these parameters do not predict conduction velocity in each situation.

T Wave of Field Potential and its Relationship with Action Potential Duration

We next measured transmembrane potentials through glass pipette microelectrodes to correlate cellular repolarization with the T wave in extracellularly recorded field potentials. Previous studies have shown that the dV/dt_{max} of the T wave (T_{up}) relates to the fastest change in membrane potential during the repolarization phase of the action potentials (25), which in NRVM is around APD_{50}. Accordingly, the T_{up} coincided with APD_{50}. The peak of the T wave (T_{peak}) coincided with APD_{90}. However, this was only evident in the monolayers exhibiting field potentials with a biphasic T wave starting negatively and ending positively, which only was the case in the minority of the monolayers. Often the field potentials showed only a positive T wave that ended before the end of the action potential (Fig. 3*A*). Figure 3*B* shows a field potential with a biphasic T wave and indicates its proposed relation with the transmembrane potential. To validate this relation, we used an NRVM in silico model. Figure 3*C* shows the simulated transmembrane potential and field potential at the same location and supports the experimental findings that suggest that T_{up} relates to APD_{50} and T_{peak} to APD_{90}, in recordings that show a biphasic T-wave morphology. Using a human model in which T_{up} coincides with APD_{90}, we investigated the frequency dependence of the relation between transmembrane potential and field potential. We increased the (slow and rapid) delayed rectifier potassium channel conductance (1.1 to 4.8 and 0.5 to 1.4 mS/cm^2, respectively) to shorten the action potential and, accordingly, the field potential. Figure 4 shows that T_{up} and T_{peak} correlated positively with APD_{50} and APD_{90} over a wide range of action potential durations, respectively. Altogether, APD_{90} correlated with the T_{peak} and APD_{50} with T_{up} in field potentials recorded from monolayers of NRVMs. In the human model, T_{up} and T_{peak} occurred more or less simultaneously and both corresponded to APD_{90}.

Current Source Underlying Local Extracellular Action Potential Formation

So far, our data have indicated that deducing action potential duration from field potentials recorded from monolayers is challenging because of the low-amplitude T wave. A recently presented alternative is the generation of a local extracellular action potential (LEAP) through high-frequency stimulation at a single electrode site (26). Figure 5*A* shows LEAPs from iPSC-derived cardiomyocytes with an atrial (left) and ventricular (right) action potential shape. The shape of the LEAP mimics that of an action potential and allows for defining durations at various percentages of repolarization. Moreover, the LEAP visualizes the fast early repolarization phase in the atrial but not the ventricular myocytes (Fig. 5*B*). The duration of the field potential (T_{peak}) is shorter than the APD in atrial but longer than the APD in ventricular myocytes. The mechanism underlying the formation of LEAPs is unclear (Fig. 5*C*). It has been assumed that high-frequency pulses cause electroporation leading temporarily to small holes in the membrane (27). Assumably, electroporation will lead to a fixed depolarization level of the membrane as ions freely diffuse through the membrane holes while the cell is still electrotonically coupled to the neighboring cardiomyocyte. It has been implied that a LEAP represents the local action potential (14). We have simulated this process by clamping the resting membrane potential of a small region under a single electrode at −55 mV (referred to as LEAP simulation). The propagation and APD distribution outside the clamped region were not affected by the LEAP simulation (Fig. 5*D*).

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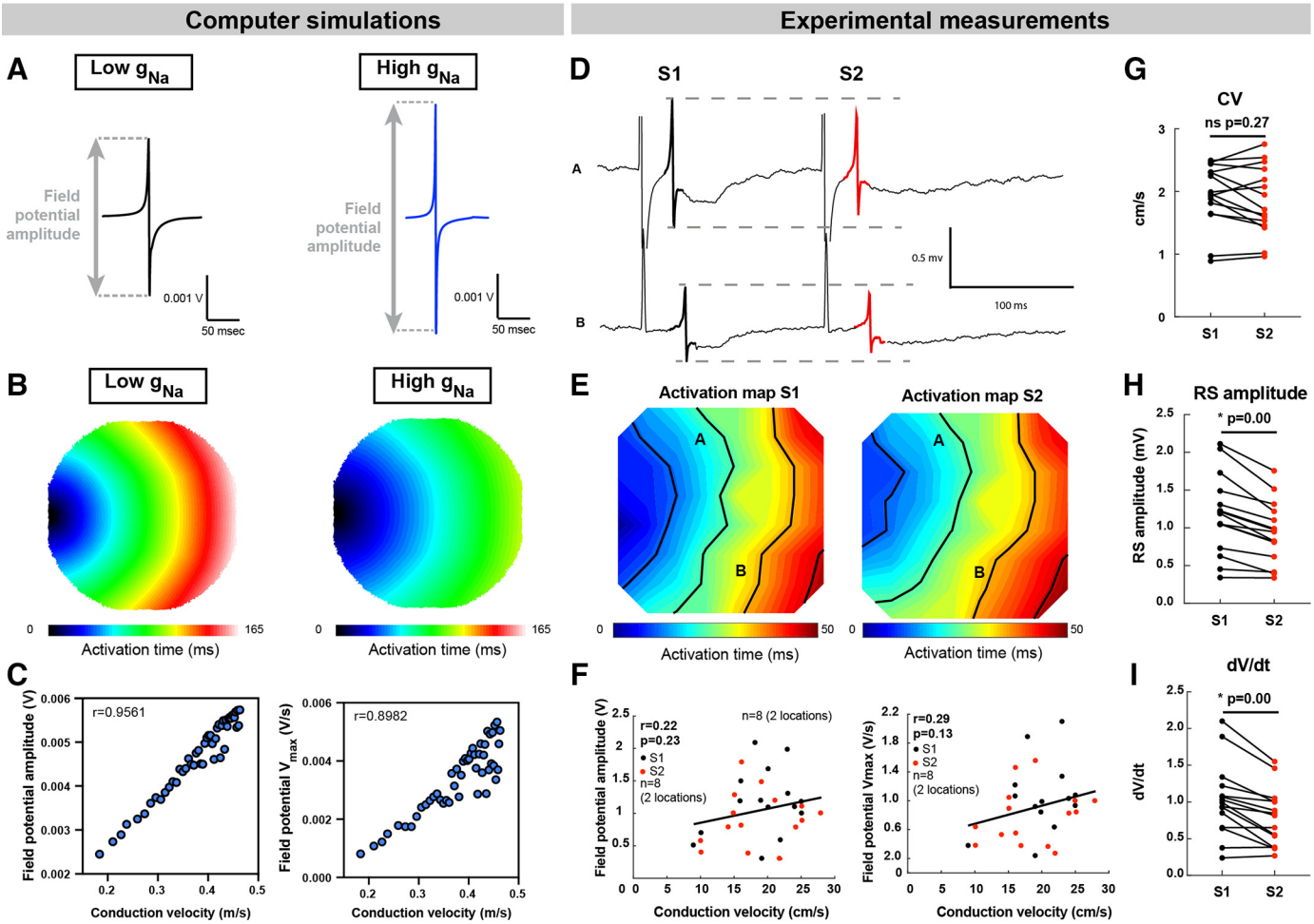


Figure 2. Field potential amplitude correlates with conduction velocity. *A*: field potentials obtained from simulations in which the fast sodium channel conductance (g_{Na}) was modified by a factor of 1.05 (*left*, low g_{Na}) and 5.15 (*right*, high g_{Na}) of baseline conditions. *B*: example of two activation maps constructed from simulations in which the fast sodium channel conductance (g_{Na}) was modified by a factor 1.05 (*left*, low g_{Na}) and 5.15 (*right*, high g_{Na}) of baseline conditions. Activation times according to the scale below. *C*: correlation between field potential amplitude (*left*), field potential maximum downslope velocity (V_{max}; *right*), and conduction velocity. Nonparametric Spearman's correlation. $r = 0.9561$, $P < 0.0001$, $n = 49$ for field potential amplitude. *D*: electrograms recorded from a neonatal rat cardiomyocyte monolayer from locations A and B as indicated in the activation maps of *E*. *F*: correlation between field potential amplitude (*left*), field potential maximum downslope velocity (V_{max}) (*right*), and conduction velocity. Nonparametric Spearman's correlation. $r = 0.22$, $P < 0.23$ for field potential amplitude. The bar graphs show local conduction velocity (*G*), RS amplitude (*H*), and dV/dt_{min} of downslope of RS complex (*I*) at locations A and B before and after premature stimulations (black vs. red dots). A paired *t* test was used to determine significance.

At the location where the membrane potential was clamped (location Z in Fig. 5*D*), we observed a transmembrane potential of −55 mV at rest and a peak voltage of −35 mV (Fig. 5*E*, *right*). At the exact location in the extracellular space, we recorded an extracellular monophasic potential resembling the shape of a LEAP (Fig. 5*E*, *right*). However, this LEAP potential differed in amplitude and shape from the transmembrane potential at that location before LEAP induction. Figure 5*F* shows the overlay of the LEAP at location Z with the transmembrane potential recorded at location Z before the LEAP simulation. The signals were scaled to obtain the same amplitude. Although both the transmembrane potential and LEAP reached full repolarization at similar moments, the shape of the repolarization phase was clearly different. The latter indicates that the LEAP is a good measure of APD_{90} but underestimates APD_{20-50}.

Our data suggest that the LEAP potential is not the equivalent of the local transmembrane potential but is caused by the current provided by surrounding cardiomyocytes. To test this, we varied the intercellular resistance in our in silico model and measured the LEAP amplitude. The amplitude of the LEAP decreased when the resistance increased (Fig. 5*A*). The amplitude of the transmembrane potential measured before LEAP induction did not change, indicating that the current causing the LEAP signal was indeed coming from neighboring myocytes. Figure 5*B* shows the superposition of transmembrane potential (black) and LEAP (red) simulations with different action potential shapes and coupling strengths. Although the terminal phase of the action potential overlaps with that of the LEAP, phases 1 and 2 often differ depending on intercellular resistance and the shape of the action potentials, further supporting the claim that LEAP inaccurately estimates APD_{20-50}.

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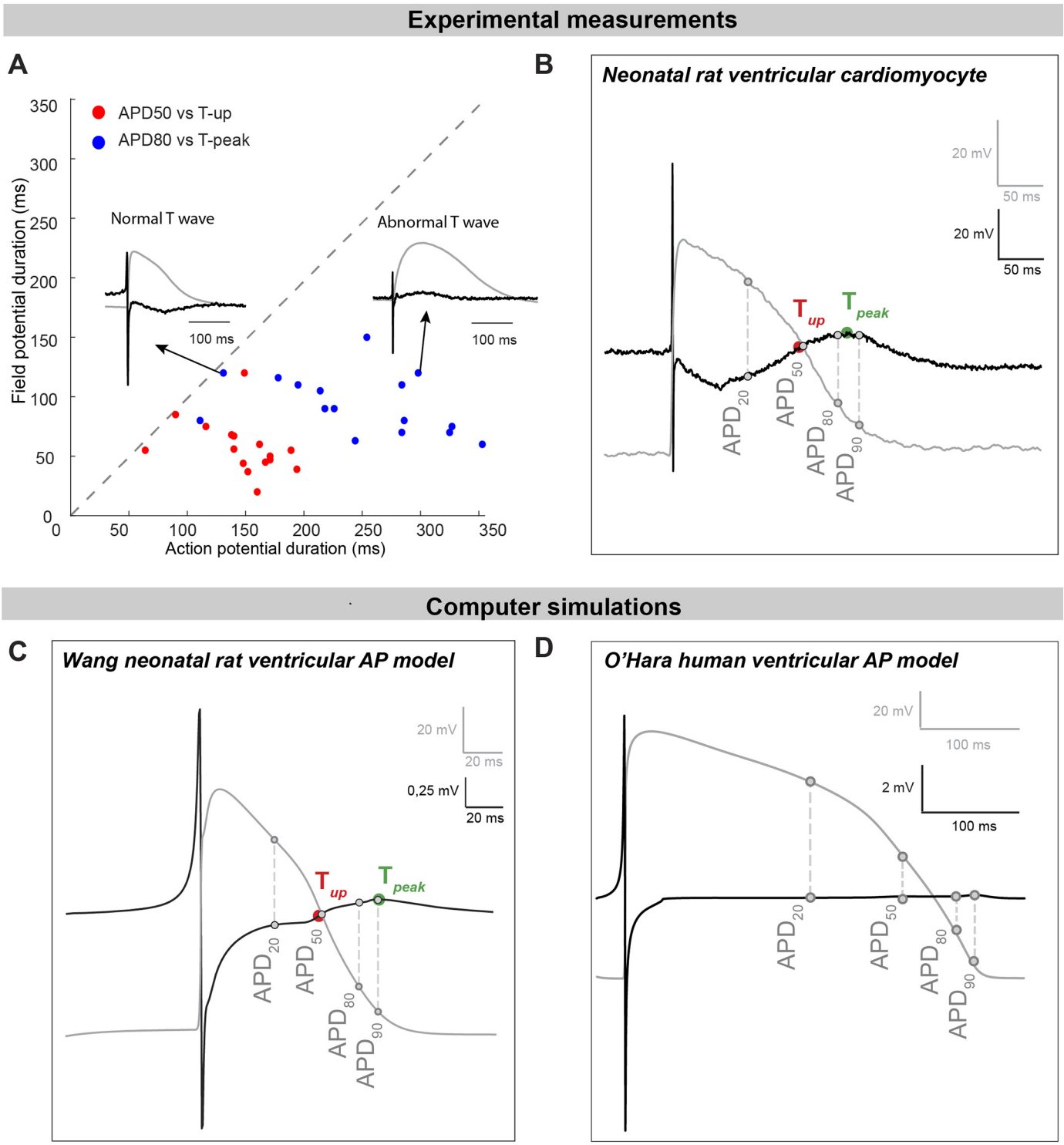


Figure 3. Comparison of field potential and action potential characteristics. *A*: scatterplot between T_{up} and APD_{50}, and T_{peak} and APD_{90} determined in field potentials and transmembrane potentials recorded using glass-pipette microelectrodes. Recordings were done in 15 monolayers at a spontaneous frequency varying between 450 and 1,000 ms intervals. Example of field potential and action potential of NRVMs obtained in vitro (*B*) and in silico using the Wang and Sobie model (*C*), and the O'Hara model (*D*). Action potential duration at 20, 50, 80, and 90% of repolarization (APD_{20}, APD_{50}, APD_{80}, and APD_{90}) are indicated on both panels. T_{up} indicates the time between activation (maximum downslope of the field potential) and maximal upstroke of the T wave. T_{peak} indicates the time between activation and peak of the T wave.

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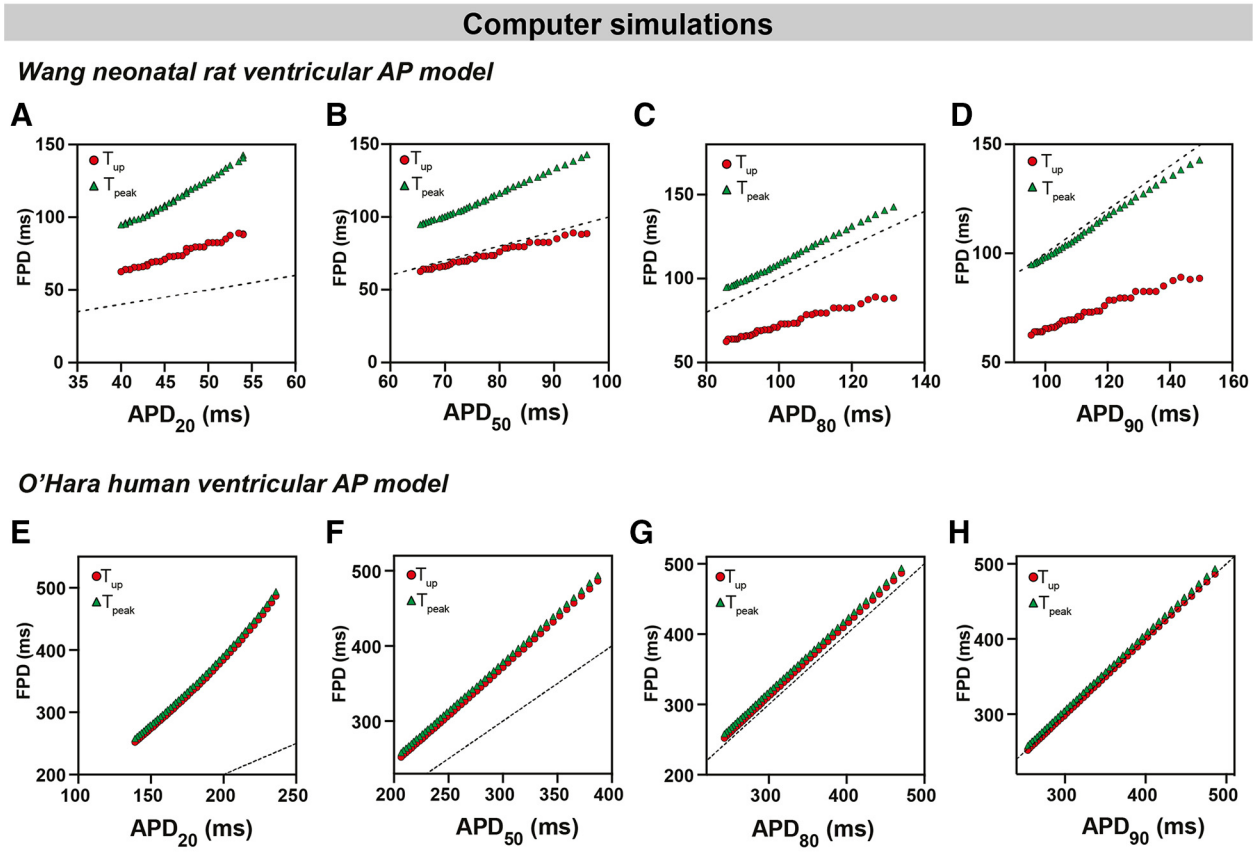


Figure 4. Field potential duration correlates with action potential duration (APD) 20, 50, 80, and 90% of repolarization. Correlation between APD_{20} (*A*), APD_{50} (*B*), APD_{80} (*C*), APD_{90} (*D*), and field potential duration (at T_{peak} or T_{up}) in the in silico Wang neonatal rat ventricular model (*A–D*) and in the O'Hara human ventricular model (*E–H*). To prolong the APD, the (slow and rapid) delayed rectifier potassium channel conductances were modified by a factor of 1.1 to 4.8 and 0.5 to 1.4 of baseline conditions in the rat and human models, respectively. T_{up} indicates the time between activation (maximum downslope of the field potential) and maximal upstroke of the T wave. T_{peak} indicates the time between activation and peak of the T wave. The dotted line indicates the line of identity. In neonatal rat ventricular myocytes (NRVMs), the APD_{90} correlates with T_{peak}, and APD_{50} correlates with T_{up}. *E–H*: example of field potential and action potential obtained from the in silico O'Hara ventricular model; APD_{20} (*E*), APD_{50} (*F*), APD_{80} (*G*), and APD_{90} (*H*) are indicated.

That the LEAP is caused by current provided by surrounding cells has implication for its use in measuring heterogeneity in repolarization. To investigate this, we induced heterogeneity in APD and conduction velocity by modifying currents in the southeast (SE), southwest (SW), and northeast (NE) quadrants, before and after LEAP simulation in the northwest quadrant (see METHODS for details). Specifically, conduction velocity was reduced in the SE and SW quadrants; AP duration was prolonged in the SE quadrant and shortened in the NE and SW quadrants (Fig. 6*A*). The field potential and transmembrane potential durations were shorter compared with the baseline (Fig. 6*D*). The overlay of the LEAP at location Z with the transmembrane potential recorded at location Z before the LEAP simulation in Fig. 6*E* illustrates that the LEAP potential deviated during all phases of the action potential from the local transmembrane potential in the presence of electrophysiological heterogeneities.

DISCUSSION

We employed an experimental and an in silico model of a monolayer of cultured NRVMs on a multielectrode array to study the interpretation of field potentials in terms of activation and repolarization characteristics. We demonstrate that the amplitude of the local RS complex of a field potential correlates with conduction velocity in silico. However, other factors influence RS amplitude in vitro, resulting in a poor correlation between RS amplitude and conduction velocity. Repolarization characteristics can be reliably derived from the field potentials of NRVMs if the T wave shows a biphasic morphology. In those cases, T_{peak} correlates well with APD_{90} and T_{up} with APD_{50}. In addition, we simulated a LEAP and determined how accurately it corresponds to the underlying action potential shape. The LEAP is a measure of APD_{90}. The local action potential differs from the LEAP during phases 1 and 2 of repolarization. This effect is more pronounced in the presence of electrophysiological heterogeneities outside the region of a LEAP recording when a LEAP differs from local transmembrane potentials in all phases of the action potential.

Field Potential Is Similar to a Unipolar Electrogram

The use of an MEA is the standardized method for recording extracellular potentials from monolayers (6). This method finds its origin in the field of neurology where these extracellular potentials are called field potentials (28–30).

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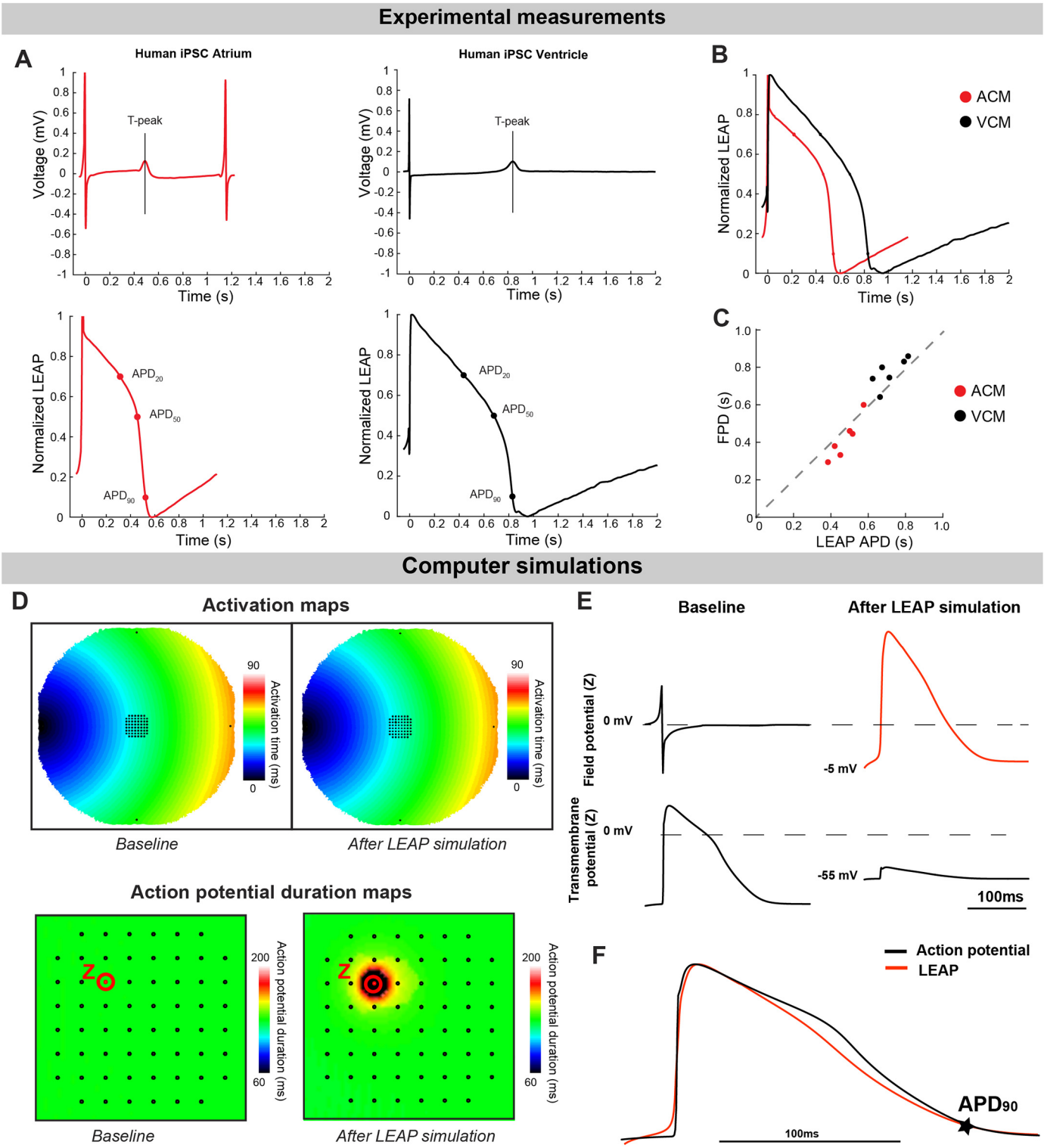


Figure 5. Local extracellular action potential (LEAP). *A*: the upper trace shows field potentials before LEAP induction recorded from a human induced pluripotent stem cell-derived (iPSC) atrial (ACM, *left*) and ventricular (VCM, *right*) cardiomyocyte. The lower traces show extracellular traces after LEAP induction at similar locations. *B*: overlay of normalized LEAP from atrial (red) and ventricular (black) iPSCs. *C*: correlation between field potential duration (T_{peak}) and APD_{90} determined in the LEAP from 12 monolayers. *D*: activation and action potential duration maps (zoomed in on electrode grid) obtained from baseline simulation and after LEAP simulation. *E*: field potentials before (*left*) and after LEAP simulation (*right*) extracted from point Z. Transmembrane potentials before (*left*) and after LEAP simulation (*right*). *F*: overlay of the transmembrane potential before LEAP simulation and the field potential after LEAP simulation.

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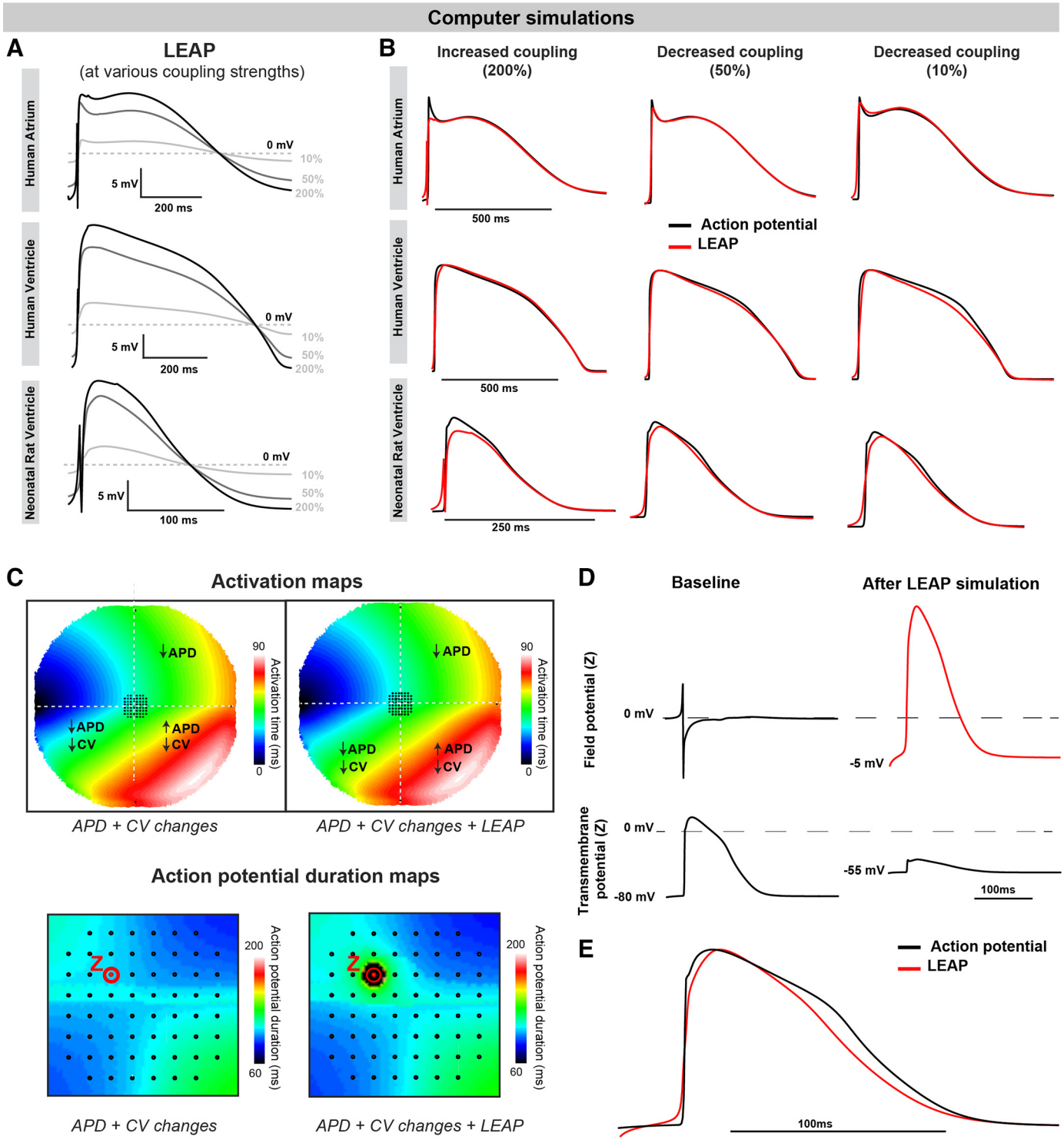


Figure 6. Local extracellular action potential (LEAP) in the presence of electrophysiological heterogeneities. *A*: LEAP simulations using a human atrial and ventricular, and neonatal rat ventricular model. Simulations were performed at 10%, 50%, and 200% of baseline intercellular coupling. *B*: overlay of transmembrane potential (black) and LEAP (red) at the aforementioned coupling strengths using the aforementioned action potential models. *C*: activation and action potential duration maps (zoomed in on electrode grid) obtained action potential duration (APD) and conduction velocity heterogeneity were induced in the monolayer by modifying currents in the southeast (SE), southwest (SW), and northeast (NE) quadrants, before and after LEAP simulation. Specifically, conduction velocity was reduced in the SE and SW quadrants; APD was prolonged in the SE quadrant and shortened in the NE and SW quadrants. Shortening or prolongation of the APD was obtained by increasing or decreasing the conductance of the inward-rectifier potassium channels (I_{Kr}) by a factor of 5 compared with the baseline. Conduction slowing was obtained by decreasing the fast sodium channel conductance (g_{Na}) by a factor of 3 compared with baseline. *D*: field potentials before (*left*) and after LEAP simulation (*right*) extracted from point Z. Transmembrane potentials before (*left*) and after LEAP simulation (*right*). *E*: overlay of the transmembrane potential before LEAP simulation and the field potential after LEAP simulation. APD, action potential duration.

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This term was adopted by the cardiac field and may have caused confusion as it gives the impression that field potentials differ from unipolar electrograms and, accordingly, require different criteria for interpretation. Efforts have been put into understanding field potential shape and providing criteria for its interpretation; the moment of local activation at the location of the recording electrode is commonly determined based on the peak of the RS complex and the local moment of repolarization is often chosen at the peak of the local T wave (15). This is not in concordance with criteria used to interpret extracellular electrograms, e.g., unipolar electrograms, which are criteria validated by experimental and in silico studies (7–11, 13). Although criteria currently used for field potential interpretation may be sufficient to describe how monolayers respond to a global slowing of conduction or prolongation of repolarization, these parameters may not be sufficiently accurate for determining the heterogeneity of activation and repolarization within the monolayer.

Interpretation of the RS Complex of Field Potentials

In an unipolar electrogram, local activation is commonly determined at the maximum downslope of the RS complex (31). Spach and colleagues (7) have validated this by simultaneously recording an action potential from the same recording site and have shown that the moment of dV/dt_{min} correlates with the dV/dt_{pos} of the upstroke of the action potential. The latter implies that the R wave preceding the dV/dt_{min} is caused by the current provided by cardiomyocytes that have depolarized earlier. Accordingly, electrograms recorded from the site of origin of activation only show a Q wave, whereas electrograms recorded from the last activated myocardium show only an R wave (13). Field potentials are similar to unipolar electrograms, which implies that dV/dt_{min} should be used to determine local activation and not the peak of the R wave as is commonly done.

Measuring the distance between electrodes and the difference in activation time between two sites allows accurate calculation of the conduction velocity (16). We demonstrate in silico that the correlation between RS amplitude and conduction velocity is high at lower values, but that the correlation is lost at more physiological conduction velocities (>30 cm/s). Sampling errors at higher conduction velocities likely cause inaccuracies in the determination of the peaks of the RS complex. In our simulations, we used a sampling frequency of 10 kHz, which is comparable to the sampling frequency used by conventional MEA systems (5, 6). For accurate determination of conduction velocity at high speeds, we recommend increasing the sampling frequency to 25 kHz.

The RS shape and amplitude have been proposed as measures for cardiac conduction velocity (32). However, our experimental recordings did not show a positive correlation between RS amplitude and dV/dt_{min} of the downslope of the RS complex with conduction velocity. We measured eight monolayers and determined local conduction velocity at two sites at baseline and during premature stimulation. We may obtain a significant correlation if we increase the number of experiments. However, that a large number of experiments are required to detect a significant correlation means that there is a lot of variation between monolayers, making the relationship between RS amplitude and dV/dt_{min} of the downslope of the RS complex an insensitive measure of conduction velocity. The variation across monolayers may be due to monolayer thickness or variations in cellular composition. In addition, RS amplitude and dV/dt_{min} of the downslope of the RS complex decreased after premature stimulation while conduction velocity did not. This also suggests that RS amplitude is an unreliable predictor for conduction velocity in vitro.

Interpretation of the T Wave of Field Potentials

Consistent with the early work of Haws, Lux, and Coronel, our data show that the time of maximum outward membrane current during local repolarization at the dV/dt_{min} of the AP corresponds to the maximum dV/dt of the T wave of the local electrogram, independent of the polarity of the T wave (10, 11). This results from an outward potassium current, which creates a positive local influence on the electrogram. In action potentials of large mammals, e.g., humans, the dV/dt_{max} of the action potential corresponds approximately with APD_{80}. Therefore, the maximum change in voltage over time of the local T wave in the unipolar electrogram aligns with the APD_{80} of the underlying action potential (25) or APD_{90}, which takes place almost at the same time as APD_{80}, as our simulations indicate. Our data indicate that the dV/dt_{max} of the local T wave (T_{up}) recorded from neonatal rat cardiomyocytes coincides with APD_{50}. This difference can be explained by the different shapes of the ventricular action potential between neonatal rats and humans. The steeper repolarization following a plateau phase in humans (phase 3 of the action potential) underlies a more pronounced T wave. In neonatal rats, the action potential is more triangular and the accompanying T wave is flatter and less pronounced. As a result, the T wave may not always be biphasic and may sometimes not be visible at all. This can result in an unreliable assessment of local repolarization as demonstrated in Fig. 3*A*. Differences in the action potential shape can be attributed to not only species differences but also differences in the developmental phase, pharmacology, or pathology (33). Our data also indicate that the T_{peak} correlates well with APD_{90} in MEAs, which supports previous findings (14, 15).

Interpretation of a LEAP

It has recently been proposed that applying high-frequency stimulation at a single MEA electrode could result in a LEAP representing the local action potential (14, 26). The LEAP is used to study the effect of drugs on the repolarization phase of the action potential, by calculating the APD_{20} to APD_{90} difference (26). Our data challenge the applicability of a LEAP for this purpose. We show that the LEAP differs in amplitude and shape from the transmembrane potential particularly during phases 1 and 2 of the action potential at the recording location, depending on the action potential shape. This indicates that the "in silico myocytes" cannot have provided the current required to generate the extracellular LEAP. The current source for the LEAP likely comes from adjacent "in silico myocytes" that have generated a full action potential. This is supported by the reduction in LEAP amplitude at higher intercellular resistance. As a result, the repolarization phase

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of the LEAP does not necessarily correspond to the repolarization phase of the local action potential at the location where the LEAP is recorded, in particular when electrophysiological heterogeneities are present (Figs. 5 and 6).

The mechanism underlying a LEAP is similar to what is known as a monophasic action potential (MAP) (11, 23, 34, 35). A MAP is an extracellular recording from a locally depolarized myocardium and is caused by the intracellular potential difference with the adjacent nondepolarized region. In humans, MAPs can be recorded by applying pressure with the tip of a recording "Franz" electrode, causing local depolarization. MAPs also occur during injury (34) or as a result of current-to-load mismatch (36).

We show that a LEAP is a record of the action potential outside the depolarized tissue (11, 23) and does not represent the local action potential. However, the duration of the LEAP is similar to the local APD_{90}, making a LEAP a preferable alternative for determining APD_{90} in the absence of a local T wave in the field potentials. A LEAP will not correctly estimate $APD_{20–50}$. In addition, it is not possible to quantify the resting membrane potential in either LEAP or MAP. Ernault and colleagues have therefore recorded transmembrane action potentials with sharp microelectrodes to measure the resting membrane potential. Although the shape of the action potential was the same under various conditions, the authors could demonstrate that a less negative resting membrane potential after exposure of the secretome of epicardial fat tissue could explain the slowing of conduction (2).

Conclusions

Local unipolar electrograms provide a strong basis for analysis of conduction velocity and repolarization characteristics in cultured monolayers of cardiomyocytes, especially at low conduction velocities and depending on the species-specific action potential shape. The T wave does not always have sufficient amplitude and in those cases, a LEAP can be used to estimate APD_{90}. LEAP does not accurately estimate $APD_{20–50}$.

DATA AVAILABILITY

Data will be made available upon reasonable request.

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AUTHOR CONTRIBUTIONS

A.C.E., R.F.M.A., H.D.D., J.R.d., R.C., E.V., and B.J.B. conceived and designed research; A.C.E., R.F.M.A., and J.L. performed experiments; A.C.E., R.F.M.A., J.L., H.D.D., J.R.d., E.V., and B.J.B. analyzed data; A.C.E., J.R.d., R.C., E.V., and B.J.B. interpreted results of experiments; A.C.E., R.F.M.A., and B.J.B. prepared figures; A.C.E., R.F.M.A., and B.J.B. drafted manuscript; A.C.E., R.F.M.A., J.L., H.D.D., J.R.d., R.C., E.V., and B.J.B. edited and revised manuscript; A.C.E., R.F.M.A., J.R.d., R.C., E.V., and B.J.B. approved final version of manuscript.

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